

Employee Attendance Data Cleaning Report

This dataset contains 300 employee attendance records collected to analyze employee attendance patterns across different departments and shifts. Each record includes Employee ID, Attendance Date, Department, Shift, Check-In, Check-Out, and Attendance Status. During data cleaning, 7 duplicate records were identified and removed, reducing the dataset to 293 records. Records with critical missing Employee IDs and Attendance Dates were then removed, resulting in 261 valid attendance records for further analysis.

Cleaning Summary

Cleaning Step	Count
Original data	300
Duplicate records removed	7
After removing duplicates	293
Missing Employee_ID removed	15
After removing missing Employee_ID	278
Missing Attendance_Date removed	17
Final cleaned records	261

Missing Value Handling Plan

Column	Missing	Method	Reason
Employee_ID	15	Delete rows	Unique identifier; cannot be reliably estimated.
Attendance_Date	17	Delete rows	Correct attendance date cannot be reliably determined.
Department	18	Mode	Categorical variable; use the most frequent category.
Shift	18	Check-In based / Mode	Use Check-In to identify shift; otherwise use mode.
Check_In	41	Median	Time value; median is less affected by outliers.
Check_Out	38	Median	Time value; median suits missing time values.
Attendance_Status	16	Logical rule / Mode	Infer from Check-In/Check-Out; otherwise use mode.

Inconsistency Standardization

Column	Inconsistency	Standardization
Department	operations / Operations	Standardize department names
Shift	MORNING / Morning	Standardize to Morning / Evening
Shift vs Check_In	Shift and Check-In sometimes do not match	Use Check-In time to verify/correct Shift
Check_In	Different time representations	Standardize to HH:MM
Check_Out	Different time representations	Standardize to HH:MM
Attendance_Status	present / Present; absent / Absent	Standardize capitalization and status labels

Cleaned Dataset Summary

Department Distribution

Department	Records
IT	62
Operations	58
Finance	50
Sales	45
Administration	35
HR	24
N/A	1
Unknown	1
operations	1

Shift Distribution

Shift	Records
Morning	186
Evening	83
MORNING	2
morning	2
N/A	1
evening	1
NULL	1
Unknown	1
EVENING	1

Attendance Status Distribution

Attendance Status	Records
Present	172
Late	52
Remote	31
Absent	19
present	2
NULL	1
absent	1